

SENTRA

Intelligent Monitoring & Protection System

1. Project Title

SENTRA

2. Project Overview

SENTRA is an intelligent surveillance and monitoring system designed to monitor motorcycles, bikes, and cars parked around a home.

Unlike a traditional CCTV system that mainly records video, SENTRA uses Artificial Intelligence to detect people and vehicles and identify predefined events around the monitored area.

The system can record important events, maintain an event history, and send notifications to the owner when a configured event is detected.

3. Real-World Problem

People commonly park their motorcycles, bikes, and cars outside or near their homes. Although CCTV cameras can record these areas, the owner cannot continuously watch the camera.

When an important event happens, the owner may only discover it later by manually checking the recorded footage.

Searching through hours of CCTV footage is also difficult and time-consuming.

4. Existing System

The existing solution is generally a conventional CCTV camera.

- Records video continuously.
- Stores the recorded footage.
- Requires the owner to manually monitor the camera.
- Requires manual searching when an incident occurs.
- Does not automatically identify important events.

5. Problem With the Existing System

The main limitation is that **the camera records the situation, but it does not intelligently understand the situation.**

For example, if someone approaches a parked motorcycle at night, the CCTV may record the event, but the owner may not know about it immediately.

6. Proposed Solution

SENTRA will use a camera and Artificial Intelligence to monitor the area around parked vehicles.

The AI will detect:

- People
- Motorcycles/Bikes
- Cars

The system will then identify predefined events, record relevant information, and notify the owner when necessary.

7. What Problem Are We Solving?

SENTRA solves the problem of **manual and continuous CCTV monitoring**.

Instead of requiring the owner to watch the camera all the time, the AI will monitor the camera feed and highlight important vehicle-related events.

***Traditional CCTV:** Camera → Record → Owner checks footage manually*

***SENTRA:** Camera → AI Detection → Event Detection → Record → Alert Owner*

8. Main Objectives

- 1 Monitor the area around parked vehicles.
- 2 Detect people, bikes, and cars using AI.
- 3 Detect predefined suspicious or important events.
- 4 Record important events with their time.
- 5 Store selected recordings/event information.
- 6 Send notifications to the owner.
- 7 Provide a web/mobile interface for viewing the camera and event history.

9. Basic System Architecture

Camera → AI Processing → Object Detection → Event Detection → Storage → Notification → Web/Mobile App

Initial Prototype

Camera → Laptop/PC → AI → Backend → Web/Mobile App

The laptop/PC will initially handle the AI processing to keep the prototype simple and affordable.

10. Expected Result

The final SENTRA prototype will provide an intelligent monitoring system that can monitor parked vehicles, detect people and vehicles, record important events, and notify the owner when predefined events occur.

11. Key Benefit

The main benefit of SENTRA is:

“The owner does not need to continuously watch CCTV footage. The AI helps identify important events automatically.”

12. Project Scope

- Home/house entrance or parking area.
- Motorcycle/bike detection.
- Car detection.
- Person detection.
- Predefined event detection.
- Event recording.
- Local storage.
- Notifications.
- Web/mobile monitoring interface.

Advanced features such as multiple cameras, advanced behavior analysis, or standalone edge hardware can be considered for future development.